

Homework 6

Alejandro De La Torre
MATH 421 — The Theory of Single Variable Calculus
Section 003
Instructor: Grace Work

October 2025

Collaboration Statement

Problem 1 (Ross Chapter 2, 9.3)

Suppose $\lim a_n = a$, $\lim b_n = b$, and $s_n = \frac{a_n^3 + 4a_n}{b_n^2 + 1}$. Prove that $\lim s_n = \frac{a^3 + 4a}{b^2 + 1}$ carefully, using the limit theorems.

Discussion. Here we essentially want to break the problem up into the individual limit components and apply the limit theorems 9.1-9.7 accordingly to show that the limit converges to $\frac{a^3+4a}{b^2+1}$.

For reference, here is what each of the 9.1-9.7 theorems say in a nutshell:

9.1 Theorem: Convergent sequences are bounded.

9.2 Theorem: If $\lim s_n = s$, then $\lim(ks_n) = k \lim s_n$.

9.3 Theorem: If $\lim s_n = s$ and $\lim t_n = t$, then $\lim(s_n + t_n) = \lim s_n + \lim t_n$.

9.4 Theorem: If $\lim s_n = s$ and $\lim t_n = t$, then $\lim(s_n t_n) = (\lim s_n)(\lim t_n)$.

9.5 Theorem: If $\lim s_n = s$, ($s \neq 0$) and $s_n \neq 0$ for all n, then $\lim \frac{1}{s_n} = \frac{1}{s}$.

9.6 Theorem: If $\lim s_n = s$, ($s \neq 0$), $s_n \neq 0$ for all n, and $\lim t_n = t$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$.

9.7 Theorem: (a) $\lim \frac{1}{n^p} = 0$ for $p > 0$.

(b) $\lim a^n = 0$ if $|a| < 1$.

(c) $\lim(n^{\frac{1}{n}}) = 1$.

(d) $\lim(a^{\frac{1}{n}}) = 1$ for $a > 0$.

Proof. Given that $\lim a_n = a$ and $\lim b_n = b$, we can then express $\lim \frac{a_n^3 + 4a_n}{b_n^2 + 1}$ as $\frac{\lim a_n^3 + \lim 4a_n}{\lim b_n^2 + \lim 1}$. We can express the result of $\lim a_n^3$ as a^3 , and $\lim b_n^2 = b^2$ by theorem 9.4. Next we can show

that $\lim 4a_n = 4a$ by theorem 9.2. Then, using the result from theorem 9.3, we can show that $\lim(a_n^3 + 4a_n) = a^3 + 4a$ and $\lim(b_n^2 + 1) = b^2 + 1$. Finally, using the result from theorem 9.6, we show that:

$$\lim s_n = \frac{\lim(a_n^3 + 4a_n)}{\lim(b_n^2 + 1)} = \frac{a^3 + 4a}{b^2 + 1}$$

for $b^2 + 1 \neq 0$ for all n . □

Problem 2 (*Ross Chapter 2, 9.9(b)*)

Suppose there exists N_0 such that $s_n \leq t_n$ for all $n > N_0$. Prove that if $\lim t_n = -\infty$, then $\lim s_n = -\infty$.

Discussion. Here we want to leverage the definition of divergence which is given in Ross 9.8 theorem, which states that:

We write $\lim s_n = -\infty$ provided for each $M < 0$ there is a number such that $n > N$ implies $s_n < M$.

The nuance here is that we are comparing a sequence diverging to a 'negative infinity' (t_n) to another sequence that we are given to suppose is 'more negative' (s_n) after a certain point (N_0 ; for all $n > N_0$). In other words, we are comparing a sequence diverging to negative infinity (t_n) with another sequence (s_n) that is eventually less than or equal to it.

So if $\lim t_n = -\infty$ is true, then we know by theorem 9.8 that for every $M < 0$ there is a number N_1 such that $n > N_1$ implies $t_n < M$. We also know that there has got to be a N_0 (we suppose there is) such that $s_n \leq t_n$ for all $n > N_0$, so we can then just define an N that is greatest of the set of $\{N_0, N_1\}$ in whatever arbitrary case we take on (which we know much exist) so that we can say $s_n \leq t_n < M$ so that we can henceforth conclude that $s_n < M$ which is what we need to say that s_n diverges.

Proof. Let $M < 0$. We know that there exists an N_1 such that $n > N_1$ implies $t_n \leq M$.

Let $N = \max\{N_0, N_1\}$. Thus, $n > N$ implies

$$s_n \leq t_n \leq M$$

which implies

$$s_n \leq M$$

Therefore $\lim s_n = -\infty$ □

Problem 3 (*Ross Chapter 2, 9.10(b)*)

Show that $\lim s_n = +\infty$ if and only if $\lim(-s_n) = -\infty$.

Discussion. We want to utilize theorem 9.8 and use a distinct M according to each case (since this is an iff statement) and prove divergence accordingly.

We will begin by show it first in the forward ($\implies$) direction: "if $\lim s_n = +\infty$, then $\lim(-s_n) = -\infty$, wherein we would let $M < 0$ so that we can say $s_n > -M$ then manipulate the expression to arrive at the conclusion that if $\lim s_n = +\infty$ then $\lim(-s_n) = -\infty$.

Then we would show the backward ($\impliedby$) direction of the iff statement such that: "if $\lim(-s_n) = -\infty$ then $\lim s_n = +\infty$ ", and follow the same procedure but with $M > 0$ to show the divergence toward a positive ∞ result.

Proof.

($\implies$) Let $\lim s_n = -\infty$. Let $M < 0$. For each $M < 0$ there is a number N such that $n > N$ implies

$$s_n > -M$$

which implies

$$-s_n < M$$

Thus, $\lim(-s_n) = -\infty$

($\impliedby$) Let $\lim(-s_n) = -\infty$. Let $M > 0$, For each $M > 0$ there us a number N such that $n > N$ implies

$$(-s_n) > -M$$

which implies

$$s_n > M$$

Thus, $\lim(s_n) = \infty$ □

Problem 4 (Ross Chapter 2, 10.2)

Prove Theorem 10.2 for bounded decreasing sequences.

Discussion. We know that Theorem 10.2 states that "All bounded monotone sequences converge", and proves such by employing the definition of $\sup S$

Proof.

Let (s_n) be a bounded decreasing sequence. Since (s_n) is bounded, the set $S = \{s_n : n \in \mathbb{N}\}$ is bounded. By the Completeness Axiom, S has an infimum; let $s = \inf S$. We will show that $\lim s_n = s$. Let $\varepsilon > 0$. Since $s = \inf S$, we know that $s + \varepsilon$ is not a lower bound for S . Therefore, there exists some $N \in \mathbb{N}$ such that $s_N < s + \varepsilon$. Since (s_n) is decreasing, we have that for all $n \geq N$,

$$s_n \leq s_N < s + \varepsilon.$$

Also, since $s = \inf S$, we have that for all $n \in \mathbb{N}$, $s \leq s_n$. Combining these two inequalities, we get that for all $n \geq N$,

$$s \leq s_n < s + \varepsilon,$$

which implies that

$$|s_n - s| < \varepsilon.$$

Thus, by the definition of the limit of a sequence, we conclude that $\lim s_n = s$.

□

Problem 5 (Ross Chapter 2, 10.5)

Prove Theorem 10.4(ii).

Discussion. Theorem 10.4 (ii) states: *If (s_n) is an unbounded decreasing sequence, then $\lim s_n = -\infty$.* In this exercise, you are asked to prove this statement directly, likely by adapting the argument used in part (i) of Theorem 10.4 for unbounded increasing sequences.

Proof.

□

Problem 6 (Ross Chapter 2, 10.7)

Let S be a bounded nonempty subset of $\mathbb{R}$ such that $\sup S$ is not in S . Prove there is a sequence (s_n) of points in S such that $\lim s_n = \sup S$. See also Exercise 11.11.

Discussion. This problem asks you to construct a sequence (s_n) within S that approaches $\sup S$. Compare this to Exercise 11.11, which states a similar result but allows S to include its supremum and requires the sequence to be increasing. In contrast, this problem only assumes $\sup S \notin S$, so the challenge is to use the definition of $\sup S$ to choose terms of the sequence that get arbitrarily close to it without ever reaching it.

Proof.

□